TOPSTECHNOLOGIES
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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

DHCP Configuration

DHCP process: Discover, Offer, Request, Acknowledge. Configure a DHCP server in Cisco Packet Tracer.

Task: Create a diagram showing the DHCP process flow.

❖ **1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.**

➢ **Aim**

- To configure a DHCP server in Cisco Packet Tracer and connect two client PCs to a network. The DHCP server will automatically assign different IP addresses to both PCs from a predefined DHCP pool.

➢ Introduction

- **DHCP (Dynamic Host Configuration Protocol)** is a network protocol used to automatically assign IP addresses and other network configuration details to devices connected to a network.
- Instead of manually configuring an IP address on every computer, a DHCP server automatically provides the required network settings to client devices.

➢ The DHCP process follows four main steps, commonly known as **DORA**:

1. **Discover:** The client searches for a DHCP server.
2. **Offer:** The DHCP server offers an available IP address.
3. **Request:** The client requests the offered IP address.
4. **Acknowledge:** The DHCP server confirms the IP address assignment.

➢ Tools and Requirements

- Cisco Packet Tracer software
- One Server-PT device
- Two PC-PT devices
- One Cisco switch (2960)
- Copper Straight-Through cables

➢ Network Topology

➢ The network consists of one DHCP server, one switch, and two client PCs.

Device	Connection	Role
DHCP Server	Switch Fa0/1	Assigns IP addresses

PC0	Switch Fa0/2	DHCP client
PC1	Switch Fa0/3	DHCP client

- **IP Addressing Plan**

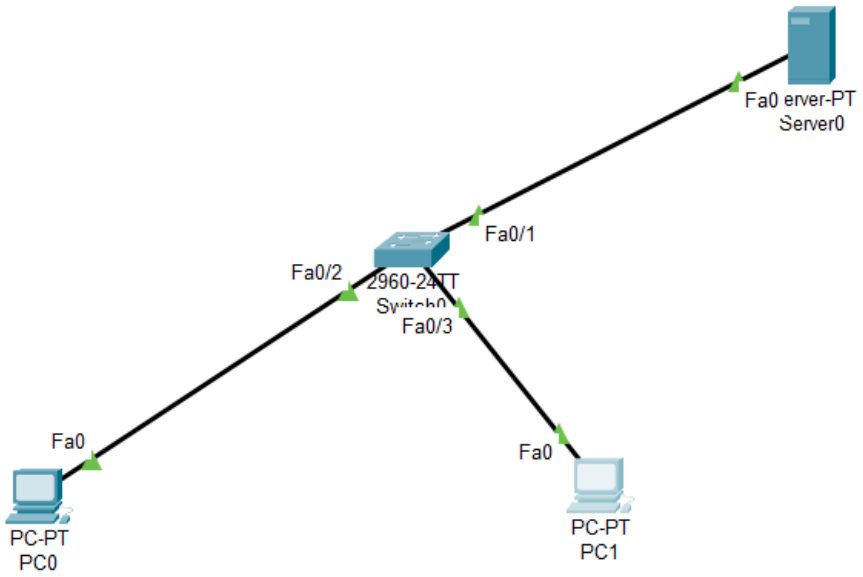
Parameter	Value
Network Address	192.168.10.0/24
Subnet Mask	255.255.255.0
DHCP Server IP	192.168.10.1
Default Gateway	192.168.10.254
DHCP Start Address	192.168.10.10
Maximum DHCP Users	50

- The DHCP server uses a static IP address, while both client PCs receive their IP addresses automatically from the DHCP pool.

- **Procedure**
- **Step 1: Create the Network Topology**

1. Open Cisco Packet Tracer.
2. Add one Server-PT device.
3. Add two PC-PT devices.
4. Add one Cisco 2960 switch.

5. Connect the server and both PCs to the switch using Copper Straight-Through cables.



➢ **Step 2: Configure the DHCP Server's Static IP Address**

1. Click the Server-PT device.
2. Open **Desktop → IP Configuration**.
3. Select **Static**.
4. Enter the following details:

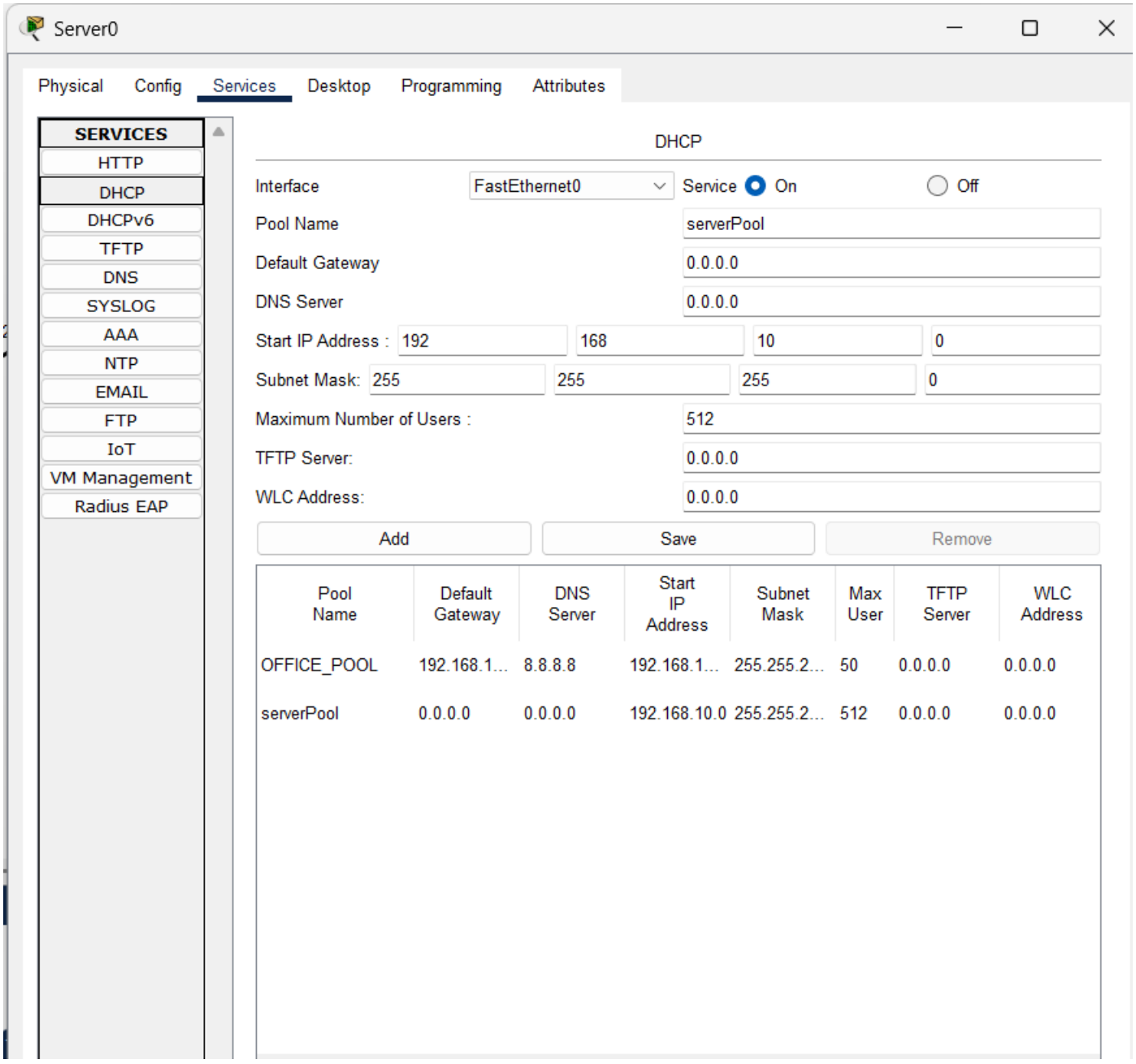
Field	Value

IP Address	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.254

- **Step 3: Enable and Configure the DHCP Service**
  1. Open the server's **Services** tab.
  2. Select **DHCP** from the left-side menu.
  3. Switch the DHCP service to **On**.
  4. Enter the following DHCP pool details:

Field	**Value**
Pool Name	OFFICE_POOL
Default Gateway	192.168.10.254
DNS Server	8.8.8.8
Start IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Maximum Number of Users	50

- Click **Add** or **Save**, depending on the Packet Tracer version.

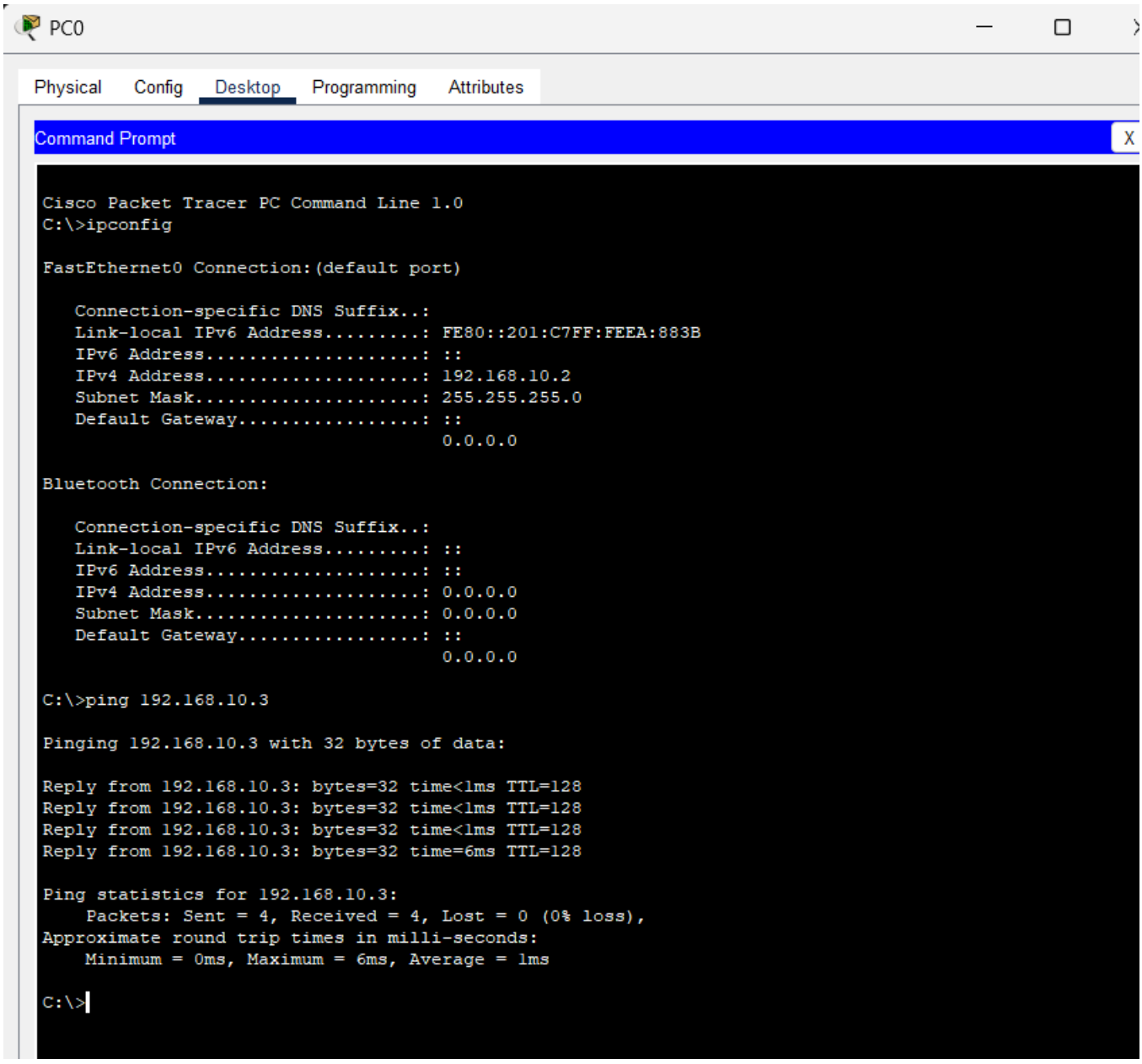


- **Step 4: Configure PC0 to Obtain an IP Address Automatically**

1. Click PC0.

2. Open **Desktop → IP Configuration**.
3. Select **DHCP**.
4. Wait for the DHCP server to assign an IP address.
5. Check the displayed IP address and subnet mask.

- **Step 5: Configure PC1 to Obtain an IP Address Automatically**
  1. Click PC1.
  2. Open **Desktop → IP Configuration**.
  3. Select **DHCP**.
  4. Wait for the DHCP server to assign an IP address.
  5. Check the displayed IP address and subnet mask.
- **Step 6: Verify the Assigned IP Addresses**
  1. Open **Desktop → Command Prompt** on PC0.
  2. Enter:
     ipconfig
  3. Note the IP address assigned to PC0.
  4. Repeat the same process on PC1.
  5. Confirm that both PCs have different IP addresses from the configured DHCP pool.
- **Step 7: Test Network Connectivity**
- Open the Command Prompt on PC0 and enter:
  ping 192.168.10.3
- Use the actual IP address assigned to PC1 if it is different. A successful ping indicates that both PCs can communicate within the same local network.


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::201:C7FF:FEEA:883B
IPv6 Address.....: ::
IPv4 Address.....: 192.168.10.2
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 6ms, Average = 1ms
C:\>

- **Observation Table**

Device	IP Configuration	Assigned IP Address	Subnet Mask
DHCP Server	Static	192.168.10.1	255.255.255.0
PC0	DHCP	192.168.10.2	255.255.255.0
PC1	DHCP	192.168.10.3	255.255.255.0

- **Result**
- A DHCP server was configured successfully in Cisco Packet Tracer. Two client PCs were configured to obtain their IP addresses automatically. Both PCs received different IP addresses from the configured DHCP pool.

❖ **2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Use arrows to show the direction of each message between client and server.**

➢ **Aim**

- To understand and represent the four stages of the DHCP process—Discover, Offer, Request, and Acknowledge—using a sequence diagram that shows the communication between a DHCP client and a DHCP server.

➢ **Introduction**

- **DHCP (Dynamic Host Configuration Protocol)** is a network protocol used to automatically assign IP addresses and other network configuration details to devices connected to a network.
- When a device connects to a network, it needs an IP address to communicate with other devices. DHCP makes this process easier by automatically providing an IP address, subnet mask, default gateway, and DNS server information.
- The DHCP process consists of four main steps, commonly known as **DORA**:
  1. Discover
  2. Offer
  3. Request
  4. Acknowledge

- **DHCP Process Sequence Diagram**
- The following sequence diagram shows how a DHCP client communicates with a DHCP server to obtain an IP address.

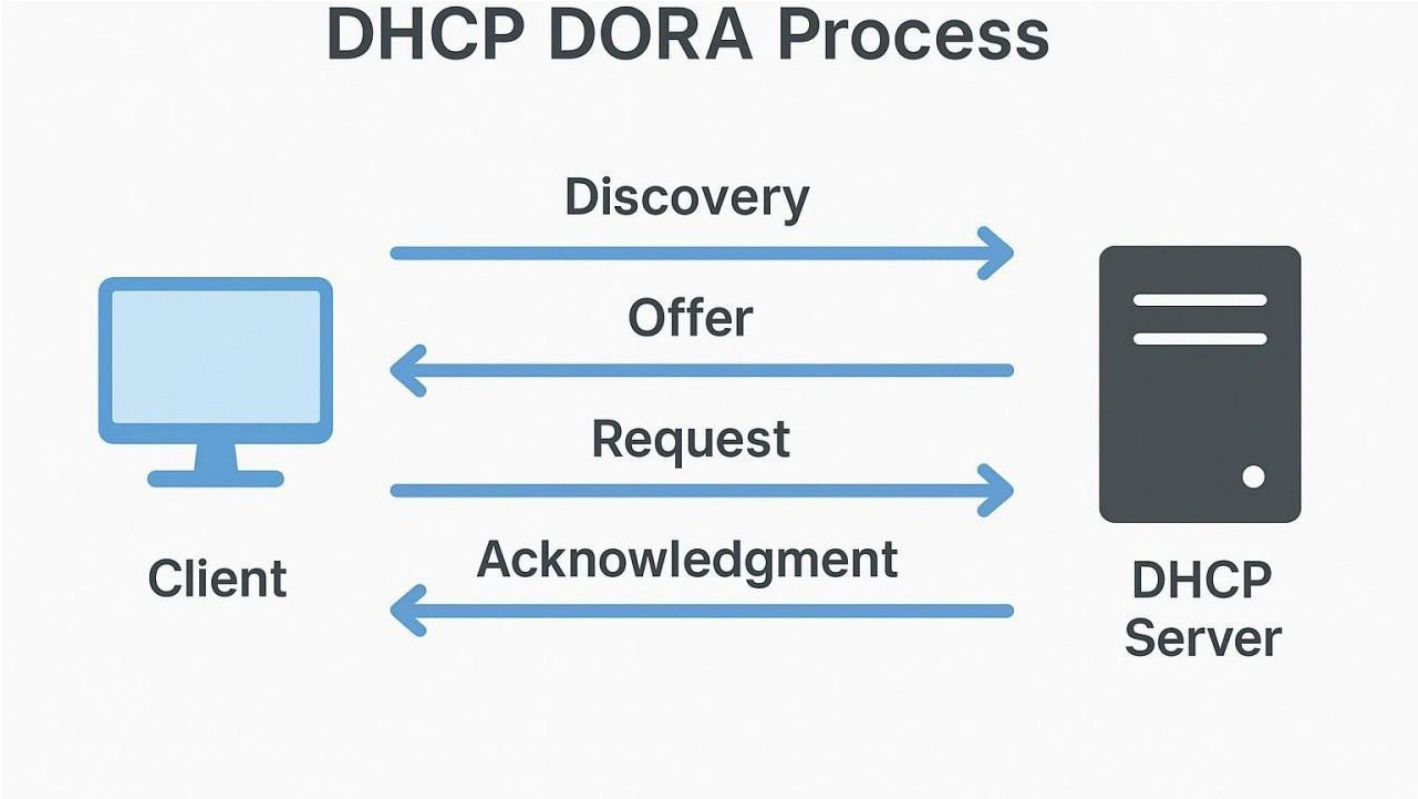


Figure 1: DHCP Process Sequence Diagram

- **Explanation of the Four DHCP Steps**
- **1. DHCP Discover**
- The client sends a DHCP Discover message to find an available DHCP server on the network. This message is generally broadcast because the client does not yet know the DHCP server's address.
  - **Sender:** DHCP Client **Receiver:** DHCP Server

- **2. DHCP Offer**
- After receiving the Discover message, the DHCP server sends a DHCP Offer message containing an available IP address and other network configuration information.
  - **Sender:** DHCP Server **Receiver:** DHCP Client

- **3. DHCP Request**
- The client sends a DHCP Request message to indicate that it wants to accept the offered IP address.
  - **Sender:** DHCP Client **Receiver:** DHCP Server

- **4. DHCP Acknowledge**
- The DHCP server sends a DHCP Acknowledgment (ACK) message to confirm the IP address assignment and provide the final network configuration.
  - **Sender:** DHCP Server **Receiver:** DHCP Client

- **Summary of DHCP Message Communication**

Step	DHCP Message	Sender	Receiver
1	Discover	DHCP Client	DHCP Server
2	Offer	DHCP Server	DHCP Client
3	Request	DHCP Client	DHCP Server
4	Acknowledge	DHCP Server	DHCP Client

- **Importance of the DHCP Process**
  - Automatically assigns IP addresses to client devices.
  - Reduces manual network configuration.
  - Prevents common IP address assignment conflicts.
  - Provides subnet mask, default gateway, and DNS information.
  - Makes connecting multiple devices to a network easier.

- **Result**
- The DHCP process sequence diagram was successfully created. It illustrates all four stages—Discover, Offer, Request, and Acknowledge—and shows the direction of communication between the DHCP client and server.

❖ **3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**

➢ **Aim**

- To limit the DHCP address pool to only one available IP address in Cisco Packet Tracer and observe what happens when two client PCs request IP addresses from the same DHCP server.

➢ **Introduction**

- **DHCP (Dynamic Host Configuration Protocol)** automatically assigns IP addresses and other network configuration details to devices connected to a network.
- A DHCP server maintains a pool of available IP addresses. When a client requests an IP address, the server assigns an available address from this pool.
- If the DHCP pool contains only one available IP address and two PCs request addresses, only one PC can receive an address at that time. The other PC cannot obtain an address until one becomes available.

➢ **Tools and Requirements**

- Cisco Packet Tracer
- One Server-PT device configured as a DHCP server
- One Cisco 2960 switch
- Two PC-PT client devices

- Copper Straight-Through cables

➢ **Network Configuration**

➢ The practical uses the following network:

Device	Role	IP Address
Server0	DHCP Server	192.168.10.1
PC0	DHCP Client 1	Assigned automatically
PC1	DHCP Client 2	Assigned automatically

➢ **DHCP Pool Settings:**

Parameter	Value
Pool Name	OFFICE_POOL
Start IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Maximum Number of Users	50

➢ Therefore, only one client IP address is available: 192.168.10.10.

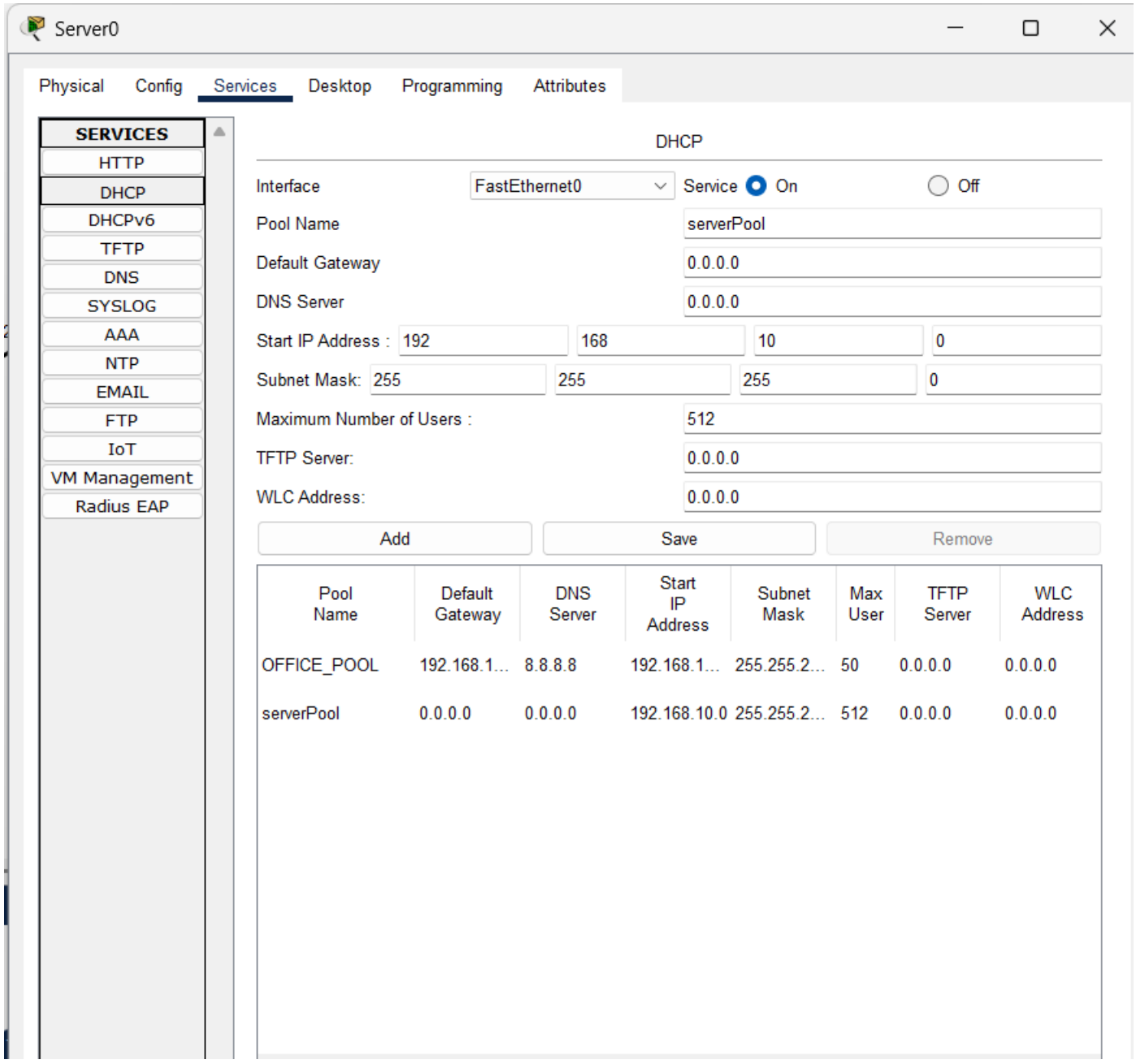
➢ **Procedure**

➢ **Step 1: Open the Existing Network**

➢ Open the previously created DHCP network in Cisco Packet Tracer containing Server0, Switch0, PC0, and PC1.

➢ **Step 2: Change the DHCP Pool Range**

- Click Server0.
- Open the **Services** tab.
- Select **DHCP**.
- Make sure DHCP Service is set to **On**.
- Select the existing pool named OFFICE_POOL.
- Set the Start IP Address to 192.168.10.10.
- Set Maximum Number of Users to 50.
- Click **Save** or **Add**, depending on the Packet Tracer version, to apply the updated pool.



- **Step 3: Configure PC0**
- Click PC0.

- Open **Desktop → IP Configuration**.
- Select **DHCP**.
- Check whether PC0 receives 192.168.10.10.
- **Step 4: Configure PC1**
- Click PC1.
- Open **Desktop → IP Configuration**.
- Select **DHCP**.
- Observe whether PC1 receives an IP address.
- **Step 5: Observe the Result**
- Check the IP Configuration window on both PCs. Only one PC should receive the single available address while the other cannot obtain an address from the pool.

- **Observation Table**

Device	DHCP Request	Expected Observation
PC0	First request	Receives 192.168.10.3
PC1	Second request	Cannot obtain an IP address

- **Explanation**
- Only one PC succeeds because the DHCP pool contains only one available IP address. Once the DHCP server assigns that address to the first PC, no free address remains for the second PC.
- **Result**
- The DHCP pool was limited to one available IP address. When two client PCs requested IP addresses, only one PC received an address from the

DHCP server, while the other could not obtain an address from the limited pool.

❖ **4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

➢ **Aim**

- To simulate a restaurant Wi-Fi network in Cisco Packet Tracer by configuring a DHCP server to automatically assign IP addresses to a laptop and a generic PC representing a smartphone.

➢ **Introduction**

- **DHCP (Dynamic Host Configuration Protocol)** is a network protocol that automatically assigns IP addresses and other network configuration details to devices connected to a network.
- In a restaurant such as Zomato-style food service environment, customers may connect their laptops and smartphones to the restaurant's Wi-Fi network. DHCP makes this process easier by automatically providing each device with a unique IP address.
- This eliminates the need for customers to manually configure their network settings.

- **Tools and Requirements**
  - Cisco Packet Tracer software
  - One Server-PT device (DHCP Server)
  - One Access Point (AP-PT)
  - One Laptop-PT device
  - One PC-PT device representing a smartphone
  - One Cisco 2960 switch
  - Copper Straight-Through cables
  - Wireless network adapters for wireless clients, where required

- **Network Topology**
- The simulated restaurant network contains one DHCP server, one switch, one wireless access point, one laptop, and one generic PC representing a smartphone.
- **Network connections:**
  - DHCP Server → Switch
  - Wireless Access Point → Switch
  - Laptop → Wireless Access Point (Wi-Fi)
  - PC representing Smartphone → Wireless Access Point (Wi-Fi)

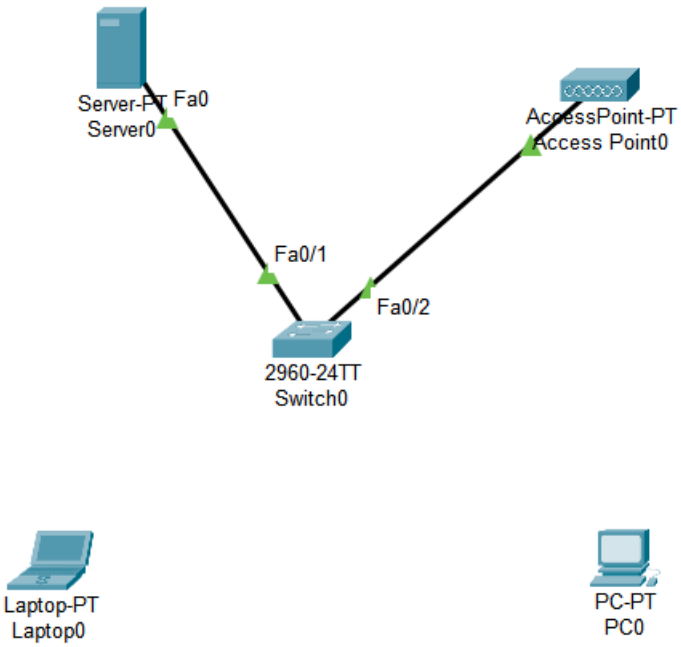
- **IP Addressing Plan**

Device	Role	IP Address
DHCP Server	Assigns IP addresses	192.168.50.2
Laptop	Wireless client	192.168.50.100

Smartphone (Generic PC)	Wireless client	192.168.50.101
Network Mask	Subnet mask	255.255.255.0

- The DHCP server uses a static IP address, while the laptop and smartphone receive their IP addresses automatically.

- **Procedure**
- **Step 1: Create the Network Topology**
- Open Cisco Packet Tracer.
- Add one Server-PT device.
- Add one Cisco 2960 switch.
- Add one Access Point (AP-PT).
- Add one Laptop-PT device.
- Add one PC-PT device to represent a smartphone.
- Arrange the devices in the workspace.



- **Step 2: Connect the Server and Access Point**
- Select Copper Straight-Through from the Connections menu.
- Connect the DHCP server's FastEthernet0 interface to Switch Fa0/1.
- Connect the Access Point's Ethernet interface to Switch Fa0/2.
- The laptop and smartphone will connect to the access point wirelessly.
- **Step 3: Configure the DHCP Server**
- Click the Server-PT device.
- Open Desktop → IP Configuration.

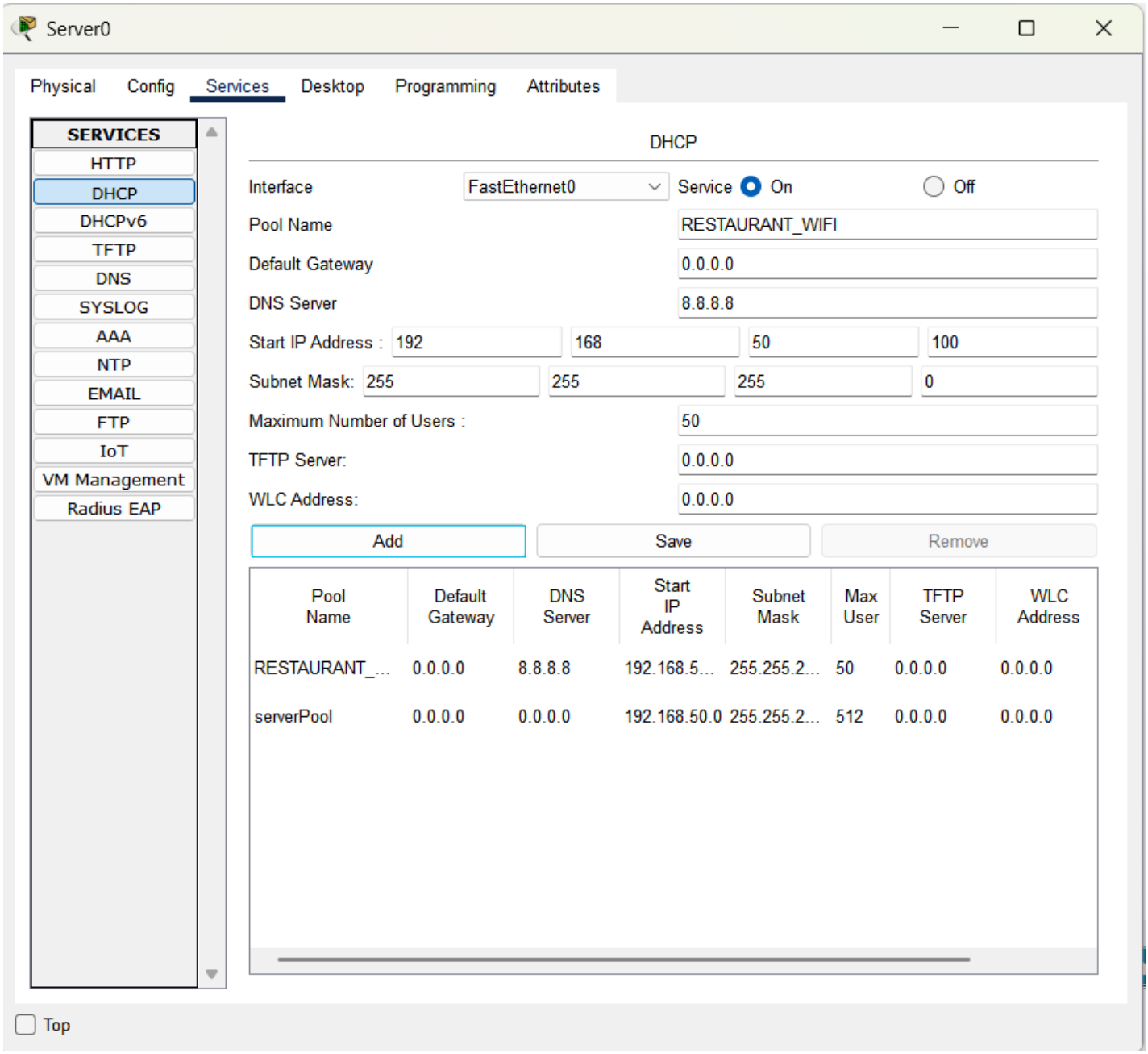
- Select Static and enter:

Field	Value
IP Address	192.168.50.2
Subnet Mask	255.255.255.0

- Open the Services tab.
- Select DHCP from the left-side menu.
- Turn the DHCP service On.
- Configure the following pool:

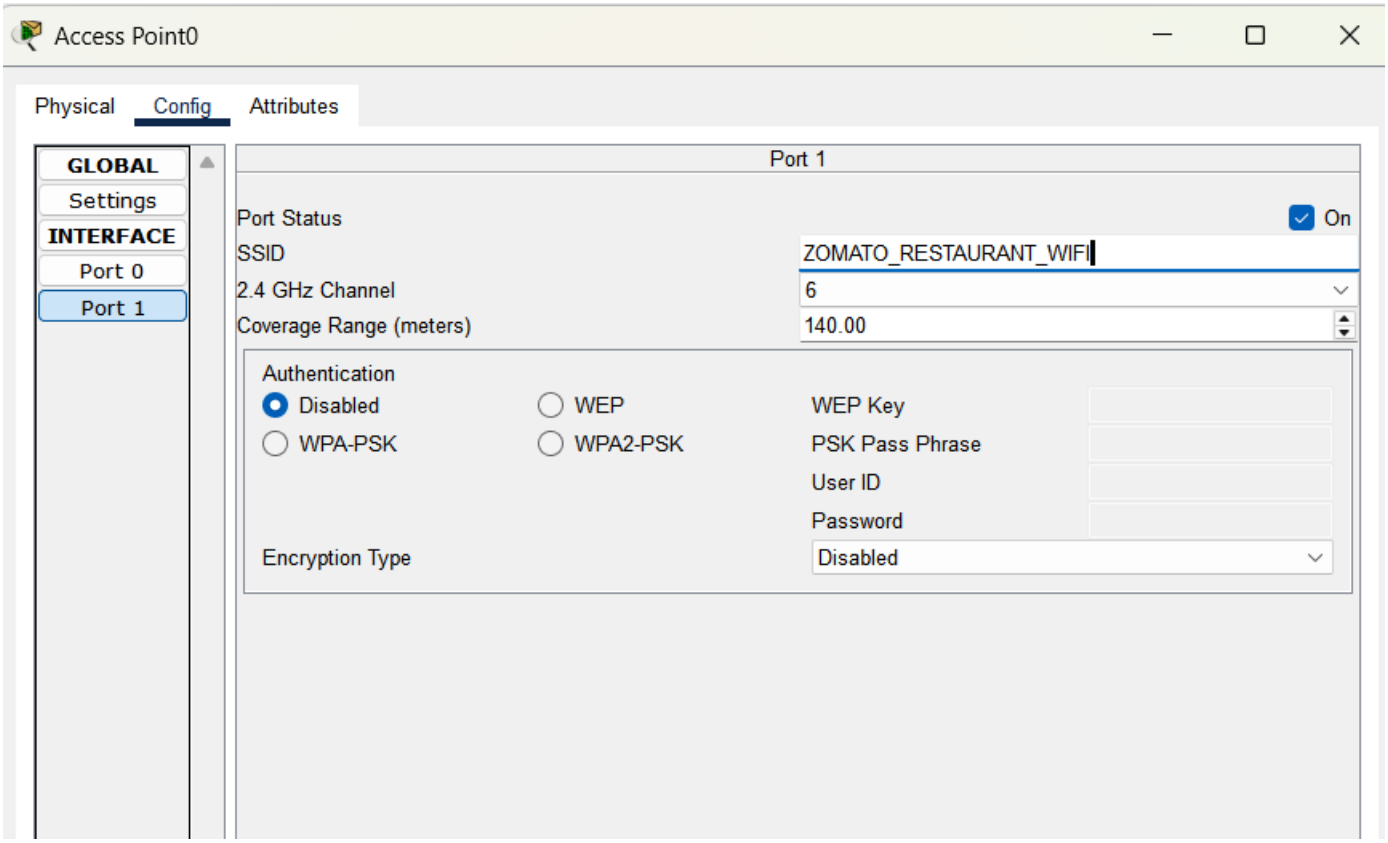
DHCP Parameter	Value
Pool Name	RESTAURANT_WIFI
Default Gateway	0.0.0.0 (not configured in this local-network demonstration)
DNS Server	8.8.8.8
Start IP Address	192.168.50.100
Subnet Mask	255.255.255.0
Maximum Number of Users	50

- Click Add or Save, depending on the Packet Tracer version.



- **Step 4: Configure the Wireless Access Point**
- Click the Access Point device.

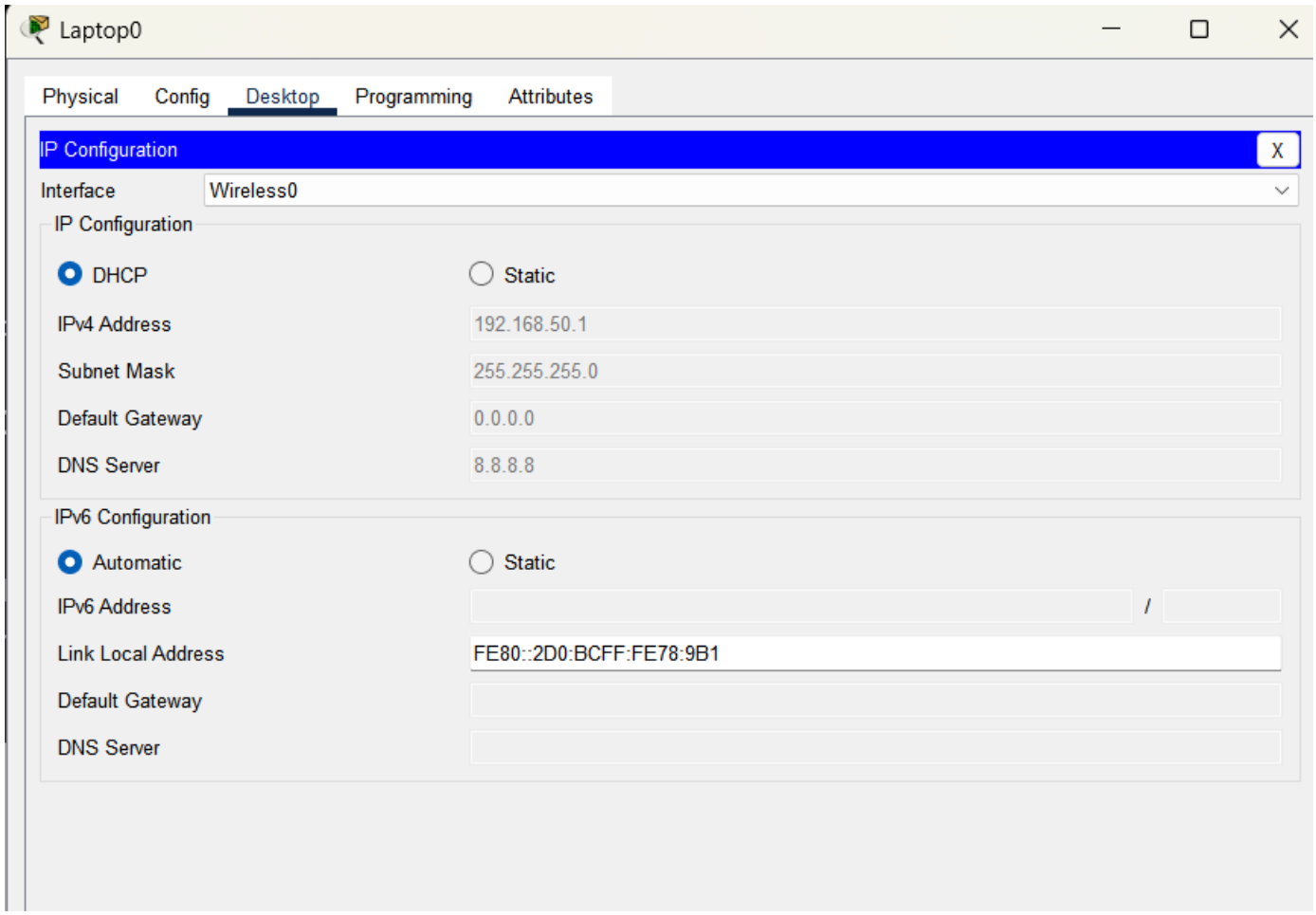
- Open its Config tab.
- Configure the wireless network name (SSID) as ZOMATO_RESTAURANT_WIFI.
- If security settings are available, select WPA2-PSK and configure a test password.
- Ensure the wireless interface is enabled.



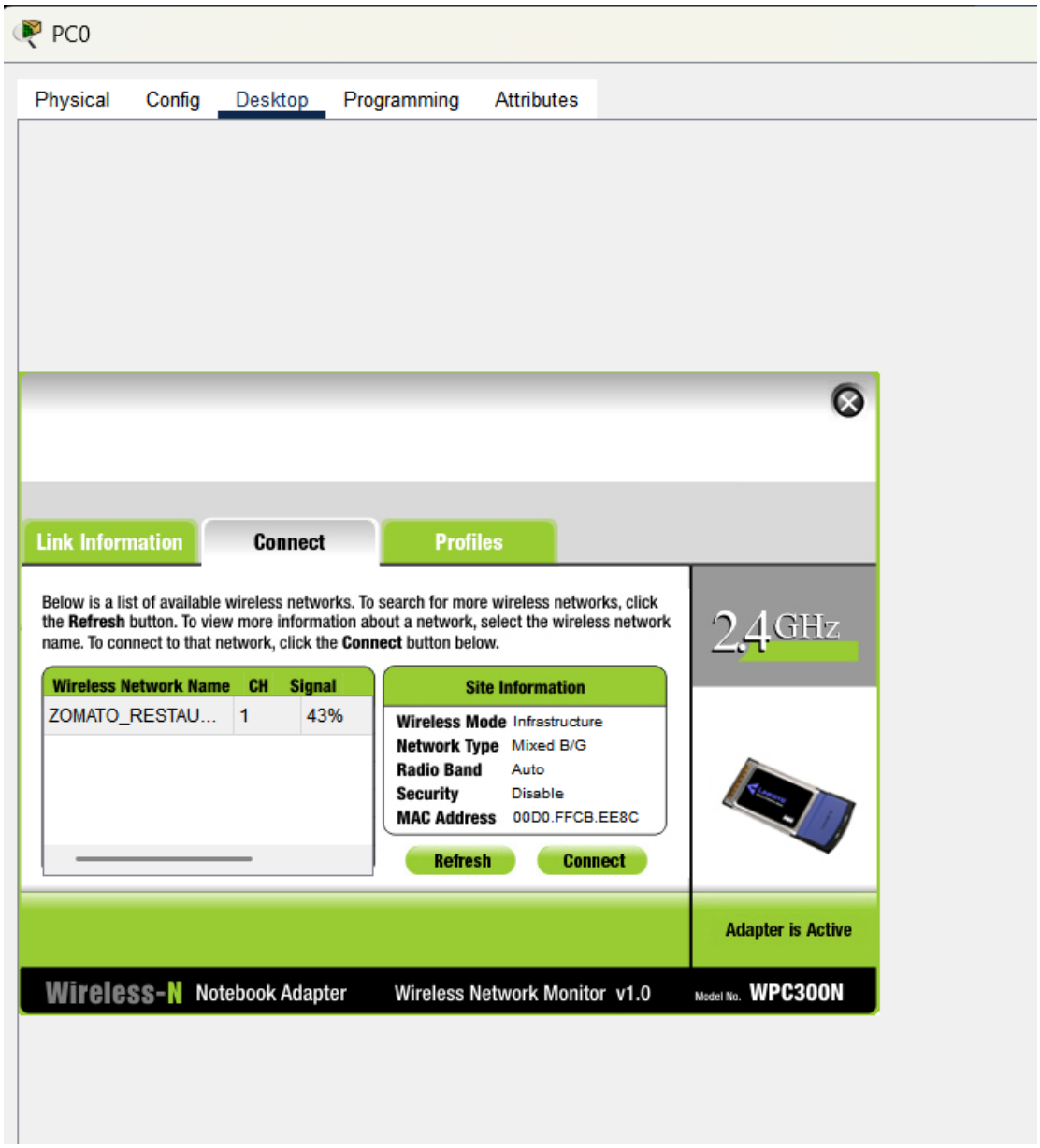
- **Step 5: Connect the Laptop to Wi-Fi**
- Click Laptop-PT.
- Open its Desktop or Config wireless settings.
- Select the wireless network ZOMATO_RESTAURANT_WIFI.

- Enter the wireless password if security is enabled.
- Open Desktop → IP Configuration.
- Select DHCP.
- Wait for the server to assign an IP address.





- **Step 6: Connect the Generic PC Representing a Smartphone**
- Click the generic PC.
- Make sure it has a compatible wireless network adapter installed and enabled.
- Connect it to ZOMATO_RESTAURANT_WIFI.
- Open Desktop → IP Configuration.
- Select DHCP.
- Wait for the server to assign an IP address.



- **Step 7: Verify the Assigned IP Addresses**
- On the laptop, open Desktop → Command Prompt.
- Enter ipconfig and note the assigned IPv4 address.

- Repeat the same process on the generic PC representing the smartphone.
- Verify that both devices have different IP addresses from the DHCP pool.

- **Observation Table**

Device	Configuration Method	Expected IP Address
DHCP Server	Static	192.168.50.2
Laptop	DHCP	192.168.50.100
Smartphone (Generic PC)	DHCP	192.168.50.101

- **Why DHCP Is Useful in a Restaurant**
  1. **Automatic IP Assignment:** Customers receive IP addresses automatically when they connect to restaurant Wi-Fi.
  2. **Saves Time:** Customers do not need to configure IP addresses manually.
  3. **Supports Multiple Devices:** The DHCP server can assign addresses to many customers using the available pool.
  4. **Reduces IP Conflicts:** DHCP manages address allocation to avoid assigning the same active address to multiple clients.
  5. **Easy Network Management:** The restaurant administrator can manage network settings centrally through the DHCP server.

- **Result**
- A restaurant Wi-Fi network was simulated successfully in Cisco Packet Tracer. A DHCP server was configured to provide IP addresses

automatically to a laptop and a generic PC representing a smartphone. Both client devices were configured to obtain their IP addresses through DHCP.

